

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-154***

“I.B. Corners 12 kV”

October 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Lark Energy Developments, the Interconnection Customer (IC), has proposed a 3.2 MW (1.2 MWC) solar generating facility to be located in Queen Anne's County, Maryland. PJM studied AC1-154 as a 3.2 MW injection into the Delmarva Power and Light Company (DPL) system at the Choptank Electric Cooperative's (CEC) I. B. Corners 69 kV Substation. The project was evaluated for compliance with reliability criteria for summer peak conditions in 2020. The planned in-service date, as requested by the IC, is November 1, 2018.

Point(s) of Interconnection

The Interconnection Customer requested a distribution level interconnection. Distribution facilities in the area of the AC1-154 project are owned by the Choptank Electric Cooperative (CEC). As a result, AC1-154 will interconnect with the CEC system at a tap of the 12 kV feeder from the I.B. Corners Substation. The DPL transmission system supplies the I. B. Corners Substation (see Attachment 1).

Transmission Owner Scope of Work

There is no DPL Attachment Facility or Network Upgrade work required for the AC1-154 project. The IC must contact CEC for their work scope and schedule.

Required Relaying and Communications

Three phase 69 kV Bus PTs and Overvoltage Protection

The AC1-154 project will require the addition of three phase potential monitoring devices on the 69 kV bus at the I.B. Corners Substation in order to sense over-voltages related to back-feeding a single phase to ground fault. This potential should be wired into the voltage input of a relay capable of detecting and tripping for overvoltage.

Over Voltage Protection

A relay capable of detecting overvoltage should remove the generation from service.

The following trip times are required:

Trip in 0.16 seconds for $V < 50\%$

Trip in 2 seconds for $50\% < V < 88\%$

Trip in 1 second for $110\% < V < 120\%$

Trip in 0.16 second for $120\% < V$

Interconnection Customer Scope of Work

The Interconnection Customer assumes full responsibility for design and construction of all facilities associated with the AC1-154 generating station and the direct connection line on the IC side of the Point of Interconnection.

The IC will be required to install metering and telemetry equipment to provide revenue metering and real-time telemetry data to PJM. The requirements for this equipment are listed in Appendix 2, Section 8 of Attachment O to the PJM Tariff, as well as PJM Manuals 01 and 14D.

It is the IC's responsibility to send the data that PJM and DPL requires directly to PJM. The IC will grant permission for PJM to send DPL the following telemetry that the IC sends to PJM: real time MW, MVAR, volts, amperes, generator status, and interval MWH and MVARH.

Summer Peak Analysis - 2020

Transmission Network Impacts

Potential transmission network impacts are as follows:

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Steady-State Voltage Requirements

None

Short Circuit

No issues identified.

Stability and Reactive Power Requirement

Not required.

Light Load Analysis - 2020

Not required.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Attachment 1

